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Video coding principles

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Outline

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Video compression principles

- Spatial redundancy
  - Images are made up of homogeneous regions
- Time redundancy
  - Successive images in a video are similar each to the other
- A video compression algorithm must exploit both kind of redundancy
- While spatial redunancy is effectively dealt with by linear transform, *prediction* is the most effective tool for removing temporal redundancy
- Temporal prediction implies computing a predictor and sending the predictor parameters along with the compressed prediction error to the decoder
- If the prediction error is *sparser* than the signal, the compression is more effective

Video compression principles

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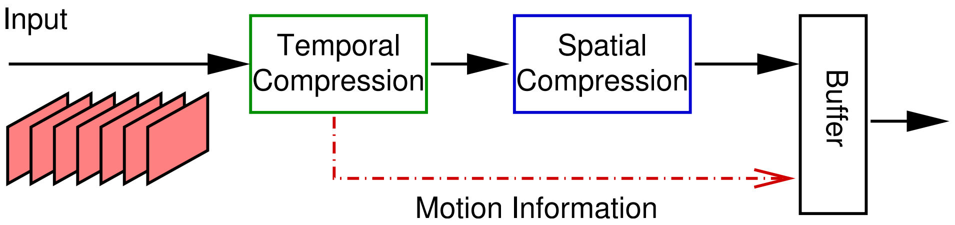
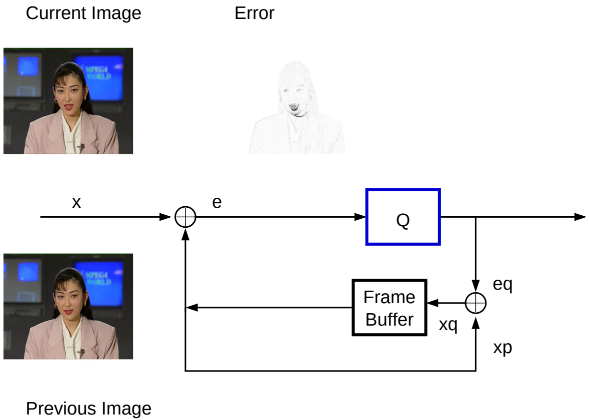
General scheme of a video coder



Prediction in video coding: DPCM

- Successive images are very similar
- Prediction:
  $\widehat{f}_{n,m,k} = \widetilde{f}_{n,m,k-1}$
- No prediction parameter to send
- What if images are different?



Block-matching motion estimation

- For each $B_k^{(\boldsymbol{p})}$, we look for the most similar block $B_h^{(\boldsymbol{p}+\boldsymbol{v})}$ in the reference image
- When we find a "match" we consider that it is the same object, which has been displaced by a vector $\boldsymbol{v}$
- The most similar block is the one that minimizes a "distance" , possibly with a regularization term

$$J(\boldsymbol{v}) = d(B_k^{(\boldsymbol{p})}, B_h^{(\boldsymbol{p}+\boldsymbol{v})}) + \lambda_{\mathrm{ME}} r(\boldsymbol{v})$$

  also referred to as a *cost function*
- The estimated vector for the block $B_k^{(\boldsymbol{p})}$ is

$$\boldsymbol{v}^*(\boldsymbol{p}) = \arg\min_{\boldsymbol{v}} J(\boldsymbol{v})$$

Performance criteria of motion estimation

- Quality of the prediction: MSE between current block and predictor
- Coding cost: number of bits needed to encode $\boldsymbol{v}^*$ — need to define the MV **encoding technique**
- Computational cost: ME time

Motion estimation: Example

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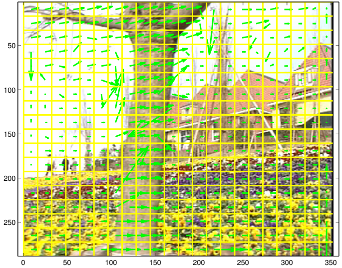
T

Reference image

Current image

Rate(MV): 1668 bits – PSNR(Pred): 24.51 dB – Time: 7.6 s



Motion Vector Components

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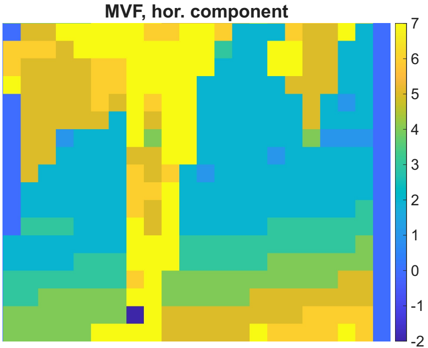
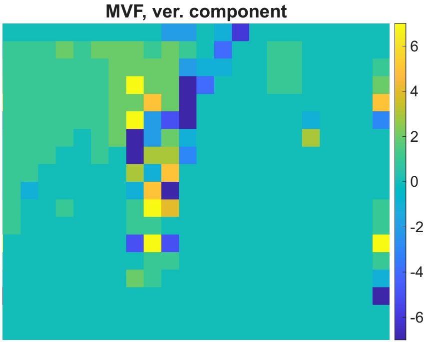
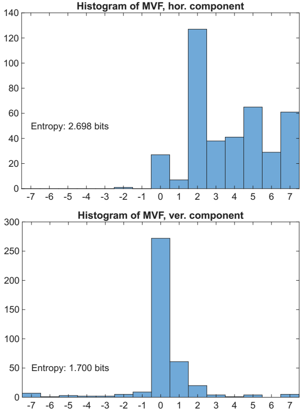




The false-color images reveal that v_x and v_y a possess a structure highly correlated with the scene geometry.

Statistical Analysis and Entropy



Histograms of MV components show a non-sparse signal.
The per-component entropy $H = -\sum p_i \log_2 p_i$ sets the theoretical limit for lossless encoding of each vector component (separately)
Direct encoding of raw MVs is inefficient due to the wide support of the probability density function

Coding Performance: Raw MV

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"Flower" sequence:

Algorithm	Total Bits	bits per vector	bits per pixel
Exp-Golomb	3048	7.697	0.0301
Huffman	1582	3.995	0.0156
Arithmetic	1570	3.965	0.0155

- Arithmetic and Huffman have close performance (H is substantially larger than 1 bit)
- EG is ineffective because distributions are not centered on zero

Predictive Coding of Motion Vectors

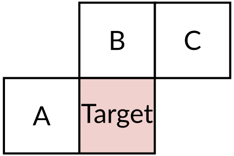
The Prediction Principle

To reduce the bitrate, we exploit the correlation between adjacent vectors. We define a Motion Vector Predictor (MVP) and encode only the Difference (MVD):

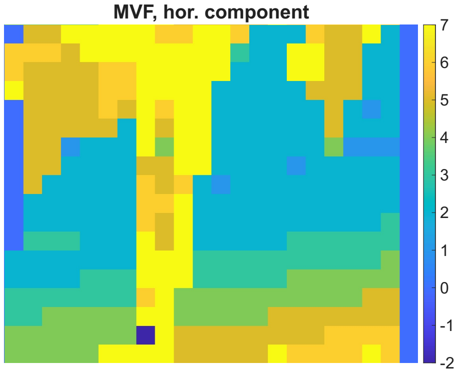
$$\boldsymbol{MVD} = \boldsymbol{MV}_{actual} - \boldsymbol{MVP}$$

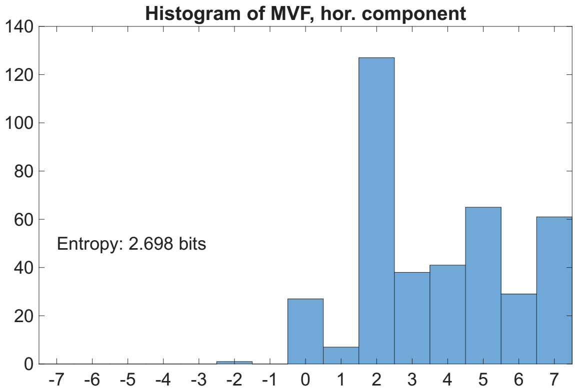
Median Predictor:

- Used in several video coding standards
- $\boldsymbol{MVP} = \text{median}(\boldsymbol{v}_A, \boldsymbol{v}_B, \boldsymbol{v}_C)$ with A: Left, B: Top, C: Top-Right.
- This ensures a robust estimate even in the presence of motion boundaries.



Statistics of the Prediction Error





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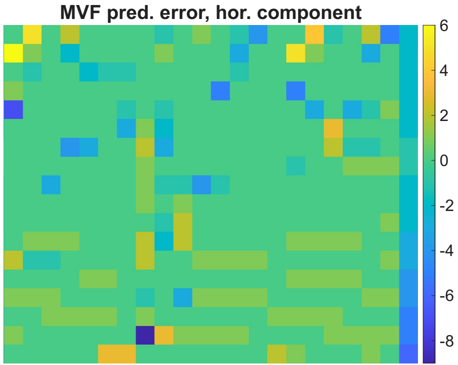
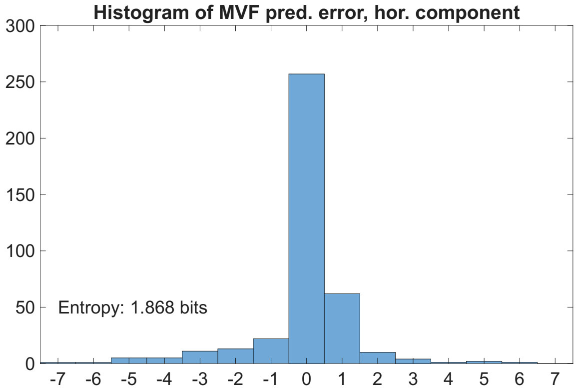
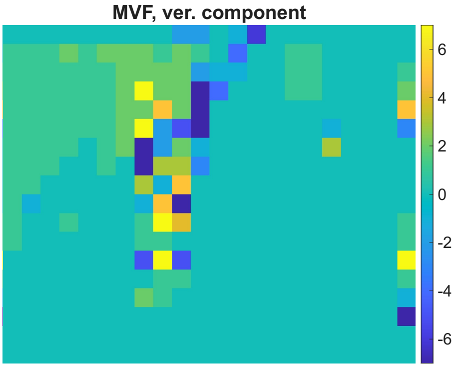
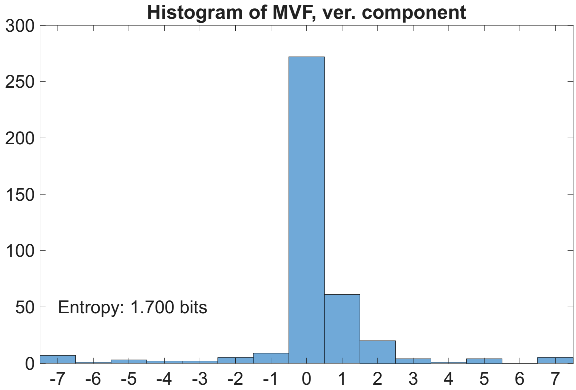
Statistics of the Prediction Error





Statistics of the Prediction Error





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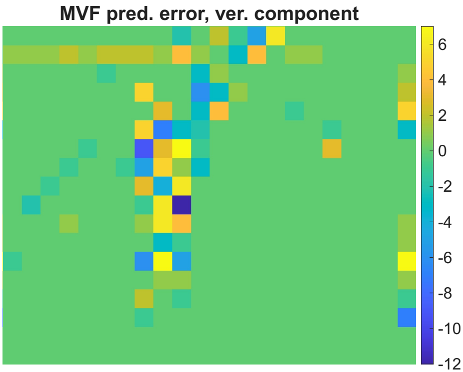

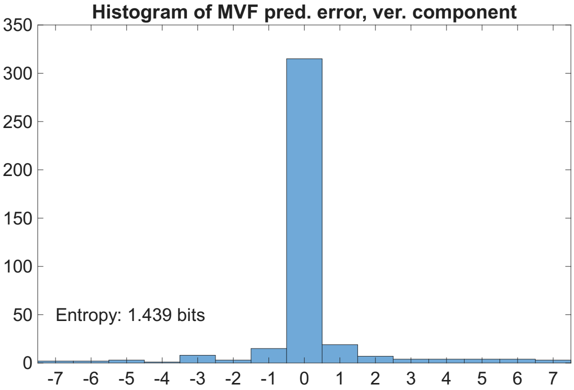

Coding Performance: Residual Vectors

Comparative Analysis

Applying spatial prediction (Median) before the entropy coder:

Algorithm	No Prediction Bits	No Prediction bpv	With Prediction Bits	With Prediction bpv	Δ%
Exp-Golomb	3048	7.70	1526	3.85	-49.9%
Huffman	1582	3.99	1211	3.05	-23.5%
Arithmetic	1570	3.96	**1199**	**3.03**	-23.6%

- **Key takeaway:** Prediction reduces substantially the MVF coding cost; the best performances are obtained by using prediction+arithmetic coding

Motion compensation

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- A prediction $\hat{I}_k$ of image I_k can be build via *motion compensation*
- The block centered in $\boldsymbol{p}$ of $\hat{I}_k$ is just a copy of a specific block in the reference image I_h
- Which block? The one pointed by $\boldsymbol{v}^*(\boldsymbol{p})$

$$\hat{I}_k(\boldsymbol{p}) = I_h(\boldsymbol{p} + \boldsymbol{v}^*(\boldsymbol{p}))$$

- This prediction is referred to as *motion compensated image*

Motion compensation

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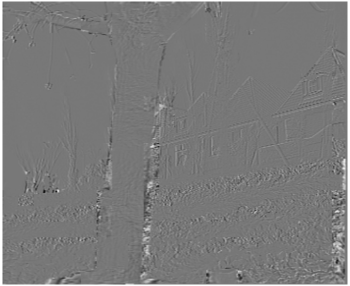

Motion-compensated image

MC error $I_k(\boldsymbol{p}) - \hat{I}_k(\boldsymbol{p})$

Motion-compensated video coding

- Video codecs work **block by block**
- Very often, the motion-compensated block is very similar to the actual block
  - In this case, it is more effective to encode the prediction error than the current block
  - Why? Because a *sparser signal* is better compressed with JPEG-like methods
- But sometimes, the prediction is not effective: new objects, disocclusions
  - In this case the encoder will encode the actual block and not the prediction error
- The encoder must signal whether it encodes the current block (*Intra coding*) or the temporal prediction error (*Inter coding*)

Summary of ME/MC: encoder

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For each block $B_k^{(\boldsymbol{p})}$ of the image k:

- We perform a ME Test with a reference h (e.g. $h = k - 1$):

$$J(\boldsymbol{v}) = d(B_k^{(\boldsymbol{p})}, B_h^{(\boldsymbol{p}+\boldsymbol{v})}) + \lambda_{\mathrm{ME}} r(\boldsymbol{v})$$

- The estimated vector corresponds to the best match:

$$\boldsymbol{v}^* = \arg \min_{\boldsymbol{v}} J(\boldsymbol{v})$$

- The encoder transmits:
  - A signalization bit (Intra: no temporal prediction/Inter: temporal prediction)
  - If Intra, it encodes $B_k^{(\boldsymbol{p})}$ with a JPEG-like method
  - If Inter, it encodes v^* and the prediction error $E(\boldsymbol{p}) = B_k^{(\boldsymbol{p})} - B_h^{(\boldsymbol{p}+\boldsymbol{v})}$ with a JPEG-like method

How to decide if a block must be encoded in Intra mode or in Inter mode? This is the **mode selection** problem

Summary of ME/MC: decoder

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For each block $B_k^{(\boldsymbol{p})}$ of the image k , the decoder reads the signalization bit:

- If *Intra* it decodes the following bits wih a JPEG-like decoder and copies the result in the block $B_k^{(\boldsymbol{p})}$ of the decoded image
- If *Inter*
  - It decodes the motion vector $\boldsymbol{v}^*$
  - It decodes the prediction error **block** $E(\boldsymbol{p})$
  - The block $B_k^{(\boldsymbol{p})}$ is built as: $E(\boldsymbol{p}) + B_h^{(\boldsymbol{p}+\boldsymbol{v}^*)}$

Design parameters of motion estimation

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- Block size and shape:
  - Small blocks: higher PSNR, higher rate, higher time
  - Variable block size: most effective even though the design is somewhat more complicated
- Cost function *d*
  - SSD optimizes PSNR, but has larger rate because of outliers
  - SAD slightly worse PSNR, slightly better rate (implicit regularization)
  - SAD+regularization: best option
- Search strategy
  - Full search: best vector, but fixed search area and very large complexity
  - Fast methods (hex, TZ): close to optimality with much less complexity
- Motion model
  - Translational: 2 parameters per block (hor and ver displacement)
  - Affine: 6 parameters per block. Highly complex, better RD performance in some cases

Motion estimation: Summary

- **Goal**: sparsify the signal
- **How**: by providing a prediction of the current block
- It is very effective and used virtually all video encoders
- Performance indicators: quality - coding cost - complexity
- Design choices:
  - Cost function (SAD, SSD, regularization, . . .)
  - Shape and size of blocks
  - Search radius and strategy
  - Motion model (translational/affine)

Outline

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The Frame types and the Group of Pictures

In all video compression methods, frames are divided into

- I Frames (Intra coded)
- P Frames (Predictive)
- B Frames B (Bi-directional or bi-predictive)

In older standards, I and P Frames are referred to as Anchor Frames (AF).
We will first describe the frame organization based on AF.

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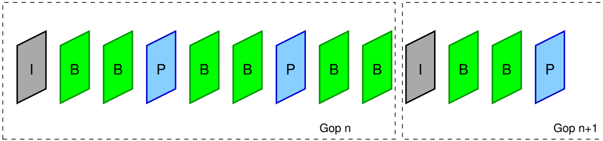
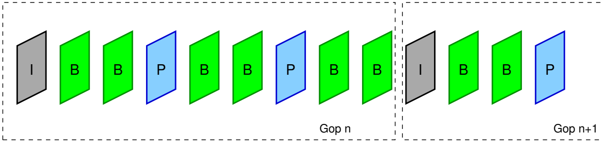
Group of Pictures

- The frames are organized into a periodical structure called **GOP** (Group of Pictures)
- The GOP defines how each frame can be predicted based on other frames
- The concept of GOP is central in all video standards
- We start by describing the first GOP architectures (MPEG-1 and MPEG-2):
  - The first frame of the GOP is **always an Intra frame**
  - Between 2 consecutive AF one finds a fixed number of B frames (possibly zeros)
  - Thus GOP structure is completely defined by:
    - the interval between I frames (GOP period)
    - the number of B frames between successive AFs



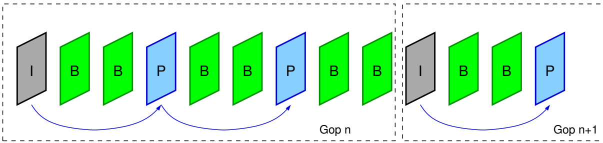
I Frames

- Encoded independently from others
- JPEG-like coding
- Low complexity, low coding rate
- Used for:
  - Random access: they can be decoded independently from others
  - Fast forward: Decoding only I-frames allows to have an accelerated video playback
  - Error robustness: They stop error propagation
- All the blocks of an Intra frame are Intra coded



P Frames

- Each P frame can be predicted from the the previous AF
- The prediction decision is taken at block level: each block of a P frame can be either Inter coded or Intra coded
- P frames have high complexity because of
- But also a much higher compression ratio than I for the same quality



B Frames

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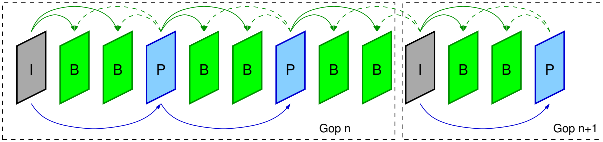
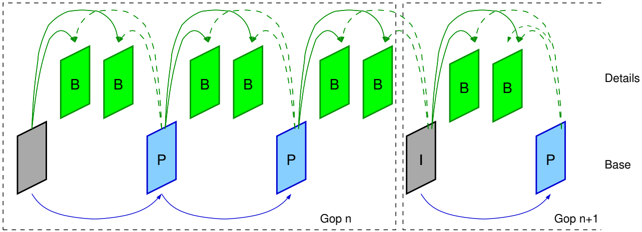
- They can be predicted from both previous and next AF
- For each block the encoder can decide if use Intra coding, forward prediction (predict from the past), backward prediction (predict from the future) or bi-directional prediction (use the average of past and future as prediction)
- Very high complexity (double ME)
- Very high compression ratio for the same quality



Frame coding order

I $\rightarrow$ AF $\rightarrow$ Frames B $\rightarrow$ AF $\rightarrow$ Frames B ...
Delay?



Rate and PSNR of different frame types

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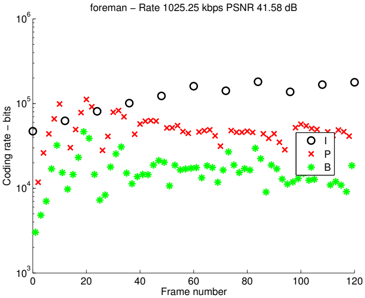

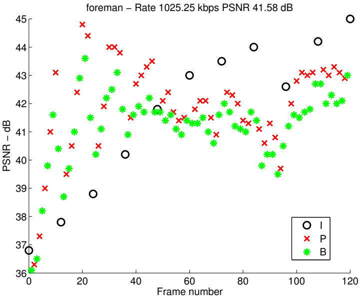

After some adjustment, we observe that I frames are approximately 3 to 5 times larger than P and 10 to 20 times larger than B

PSNR is more variable and depends on content, but the quality of I frames is typically forced to be high because they are used for predicting the whole GOP

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The hybrid video encoder

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- All practical video codecs work on the basis of blocks of pixels
- These blocks are referred to as **macroblocks** (older standards until H.264) or **coding units** (from H.265 on)
- Each MB (or CU) can be coded using one among several *coding modes*
- The set of available coding modes depends on the frame type

Intra:	No temporal prediction, transform-based coding; available for all frames
Inter:	ME/MC-based temporal prediction, transform coding; available only non-Intra frames
Direct:	Motion vector inferred from neighbors; no residual coding; available only non-Intra frames
Lossless	available on all frames

- Each mode has different rate and distortion, how to choose? This is the **mode selection problem**
- Each macroblock or coding unit can be split in smaller blocks: how to choose? This is the **block partition problem**

Coding mode & partition selection

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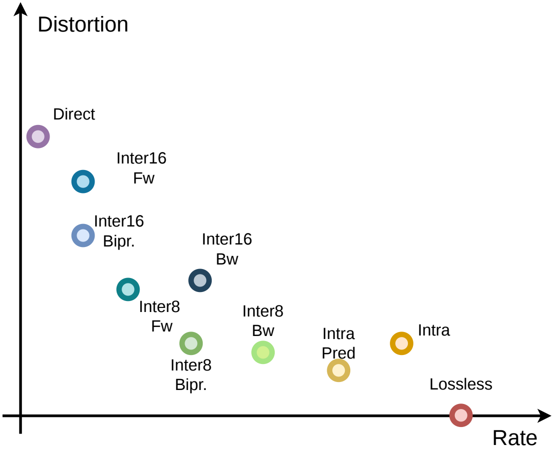

Example of RD operation points in a generic hybrid video encoder

Example. A single 16×16 block in a B frame can be encoded in many different *modes*.
Direct mode has very small bit-rate but large distortion.
Inter mode can use backward, forward and bidirectional prediction; or, the block can be split in four 8×8 blocks.
Intra mode can use or not the spatial prediction.
Finally the block can be **losslessly** encoded.
Each choice corresponds to a different (R, D) operation point.

Coding mode selection

- Let us first consider the case of fixed block-size
- Let K be the total number of blocks in a video sequence
- Goal of coding mode selection: for each block k, find the coding mode i_k such that D is minimized for a given total rate R:

$$D = \sum_{k=1}^{K} D_k(i_k, Q) \qquad\qquad R = \sum_{k=1}^{K} R_k(i_k, Q)$$

- The quantization step Q is given
- The set of modes $\boldsymbol{i} = \{i_k\}_{k=1}^{K}$ must be chosen such that we minimize:

$$J(\boldsymbol{i}, Q, \lambda) = \sum_{k=1}^{K} D_k(i_k, Q) + \lambda \sum_{k=1}^{K} R_k(i_k, Q)$$

Coding mode selection

- Joint minimization over $\boldsymbol{i}$ is way too complex
- A sub-optimal (block-wise) minimization is preferable
- For a block k, we choose the mode such that we minimize:

$$J_k(i_k, Q, \lambda) = D_k(i_k, Q) + \lambda R_k(i_k, Q)$$

- That is, we minimize *separately* the coding mode of each block
- The selected mode depends on Q and λ

Coding mode selection

- The quantization step Q is considered as an *input*
- For each Q there exists an optimal λ value, which is determined empirically for each different video codec. Examples:
  - MPEG-2 : $\lambda = aQ^2 + b$
  - H.264 : $\lambda = c2^{dQ+e}$
- The Lagrangian multiplier in motion estimation, λ_{ME}, is empirically set as the square root of the mode-selection Lagrangian multiplier: $\lambda_{\mathrm{ME}} = \sqrt{\lambda}$
- With a fixed λ, minimizing J_k amounts to find a line in the RD plane

Block partition

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- Modern video codec allow blocks B with variable size, called **coding units** (CU) with sizes from 128×128 to 4×4 (exact values depend on the standard)
- How to chose the best block size? It is the block partition problem
- This problem can be cast as a mode selection problem: splitting a block into four sub-blocks is seen as a new coding mode, with a new RD cost $J_{\mathrm{split}} = \sum_i J_{\mathrm{subblock}\ i}$
- The block partition is decided in a recursive way: we start with the largest block size and if $J_{\mathrm{split}} < J_{\mathrm{B}}$, the block is split. In this case, for each subblock we run again the algorithm to decide a possible further split
- This approach avoids to explore all the possible combinations of split decision: is a block has a better coding cost than its four subblocks, deeper partitions are not explored (suboptimal but reasonable)

Coding mode selection

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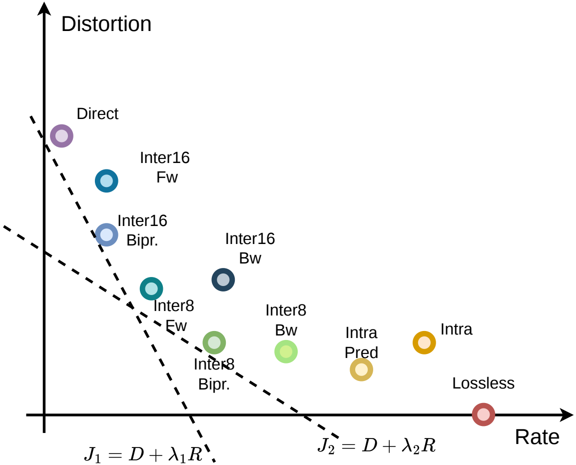


Interpretation of the Slope ($-\lambda$):

- The equation $D = -\lambda R + J$ defines a family of lines in the R-D plane with a negative slope equal to $-\lambda$.
- **High λ (Steep lines):** The encoder is constrained by a low bit budget; it will choose points closer to the y-axis (e.g., *Direct* mode).
- **Low λ (Flat lines):** The encoder prioritizes visual fidelity; it will choose points closer to the x-axis (e.g., *Inter8 Bipr.*).

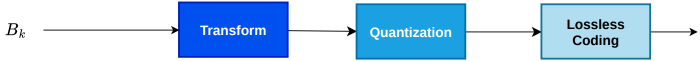
Optimization Goal: Selecting the optimal mode corresponds to finding the point that is first "touched" by the line as it moves from the origin towards the convex hull of all available coding modes.

Outline

The hybrid video encoder

- Now we have all the tools for building an hybrid video encoder and decoder
- Each image is split into blocks
- For each block, we can compute a prediction
  - **Null predictor**: the block is directly encoded (DCT+Quantization)
  - **Temporal prediction**: via ME/MC (simple or bi-prediction)
  - **Space** or **Intra** prediction: a block can be predicted from already encoded blocks in the same image
- The prediction error is JPEG-like compressed (DCT+Quantization)
- The block is the decoded and stoder, because it might serve for further predictions (temporal or spatial)
- Mode decision, Block partition, motion vectors and quantized coefficients are encoded with a VLC

The hybrid video encoder



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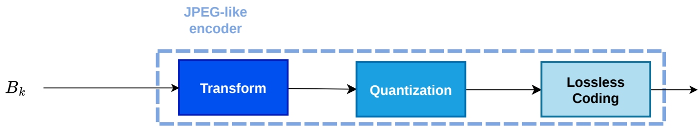

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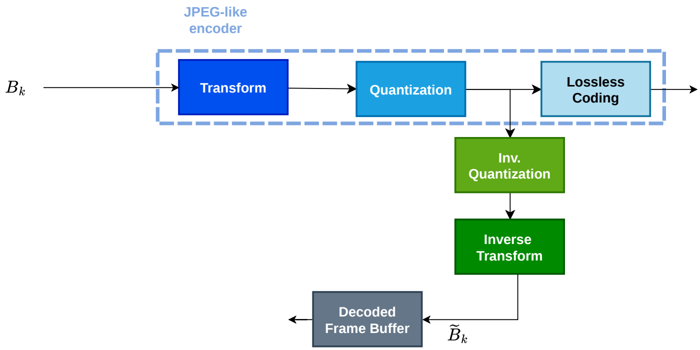

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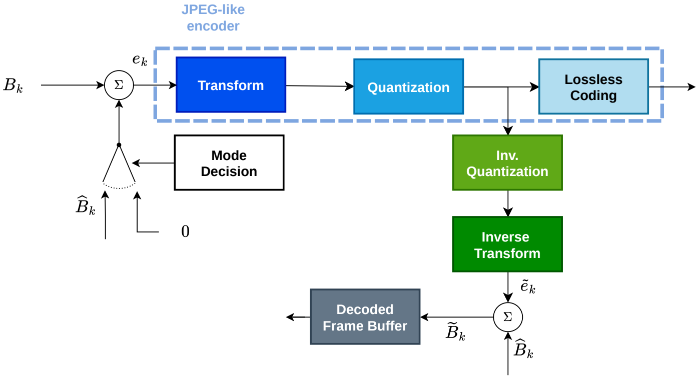

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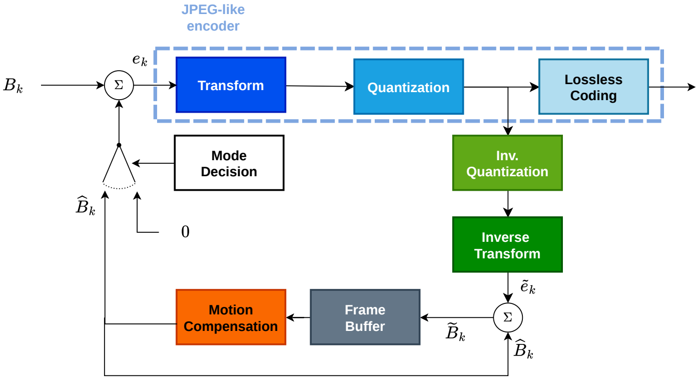

The hybrid video encoder

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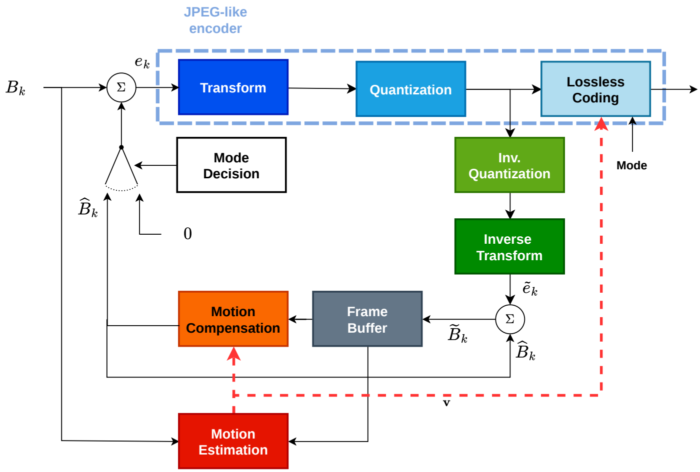

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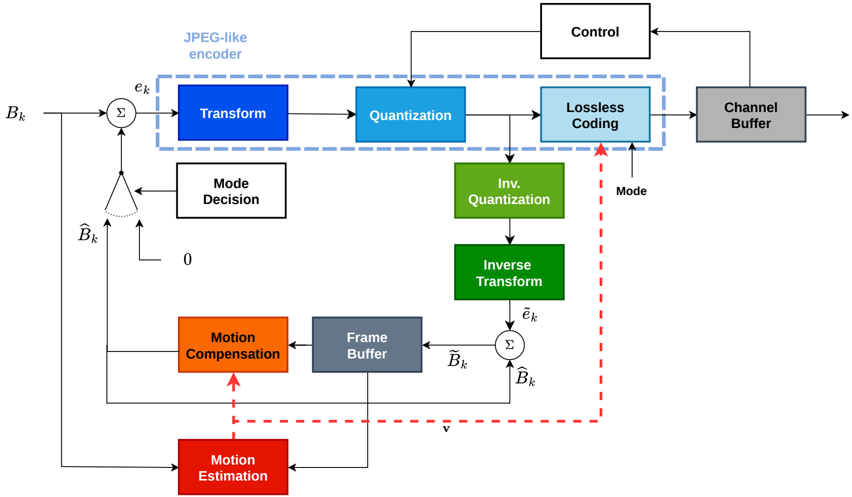
The hybrid video encoder



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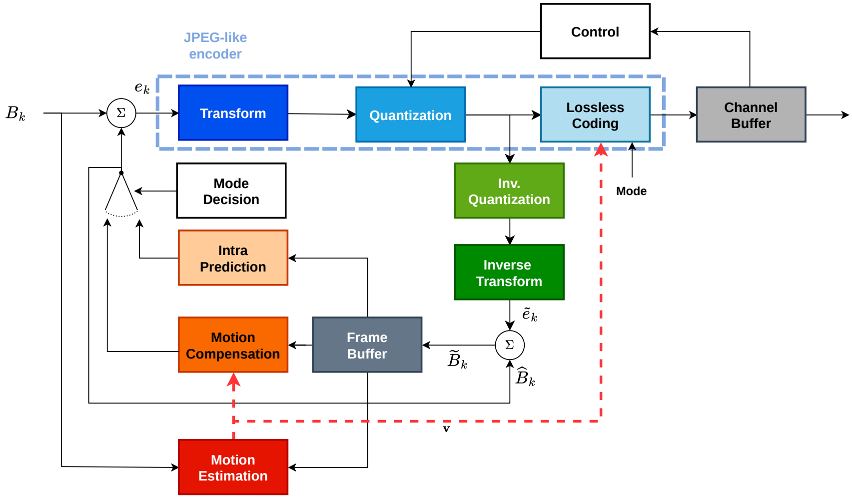
The hybrid video encoder



Rate control

- The coding rate is determined by the quantization step: this defines the rate R_C at which the compressed stream is *produced*
- But the R_C resulting from a given step depends on the content
- In many applications it is desirable to achieve a given target rate R_T
- The target rate can be related to physical constraints (e.g., the speed of a reading device) or to the network conditions (the channel capacity C)
- In order to make R_C match the reading rate R_T, a buffer (named channel buffer) and a closed loop controller are used
- The encoded video is written in the buffer at speed R_C and is read at speed R_T
- As long as $R_C > R_T$ the buffer occupancy grows; if it surpasses a threshold γ_{high}, the controller increases the quantization step, reducing R_C
- If $R_C < R_T$, the buffer empties; if it is below a second threshold γ_{low}, the controller reduces the quantization step, increasing R_C

The hybrid video decoder

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- **The decoder is a subset of the encoder**
- As we will see, standards define only the decoder operation, such to guarantee the interoperability and competitions among encoders
- First, the decoder decodes the so-called side information: the partition information (if variable block size is admitted), the coding mode (if multiple modes are possible) and the motion vector(s) (if the coding mode is based on temporal prediction)
  - This information is in the bitstream, thus the decoder does not perform mode decision, partition and motion estimation
  - These operations are computationally heavy: thus the decoder is typically much faster than the encoder
  - Since only the decoder is standardized, motion estimation, mode selection and block partition are not standardized, allowing for competition of efficient methods (e.g., hex-search ME, or fast mode decision algorithms)

The hybrid video decoder

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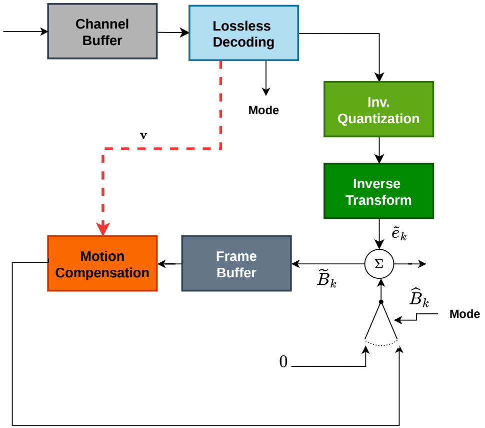

The decoder is a subset of the encoder
No ME, no mode decision, no block partition: much lower complexity

The hybrid video decoder

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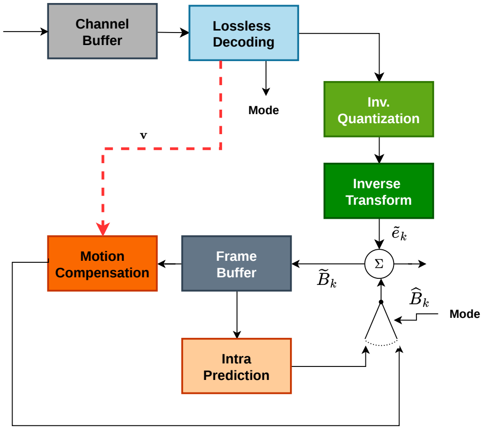

The decoder is a subset of the encoder
No ME, no mode decision, no block partition: much lower complexity

Hybrid video coding: wrap-up

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- Each frame in the sequence is labeled as I, P, or B according to the GOP structure
- The encoder works on a block-by-block basis
- For each block the encoder can choose one *coding mode* within a set that depends upon the frame type
  - For example, no motion-compensated mode can be used in I frames
- The *coding mode decision* strategy is not standardized
- The **mode syntax** is standardized
  - This means that an encoder can take a suboptimal mode decision and the encoded stream will still be decodable
  - It can make sense if the goal is to reduce the compuation time
- The optimal RD decision (dubbed as RDO) is the one minimizing $J = D + \lambda R$

Hybrid video coding: wrap-up

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- In **INTRA coding mode**, only information from the current frame can be used to encoded the block
  - Spatial coding tools can be used: linear transform (DCT) and possibly spatial prediction
- In **INTER coding mode**, information from decoded frames *other than the current one* in the DFB can be used:
  - Temporal prediction for P frames
  - Bi-prediction from B frames (information from two frames in the DFB)
  - ME parameters have a big impact on the rate-distortion-complexity trade-off
  - **Motion vector must be encoded losslessly**:
    - Huffman, Arithmetic or Exp-Golomb coding
    - Since motion vector fields typically have spatial coherence, predictive coding is often used
- The channel buffer allows to adapt coding rate to the rate of the output channel or of the storage medium
  - When the channel buffer size is above a high threshold, the quantization parameter is increased (more compression)
  - When the channel buffes size is below a low threshold, the quantization parameters is reduces (less compression)

Hybrid video coding: concept map

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